

Supplementary Material

REM sleep EEG slowing reflects brain cholinergic denervation in aging and Mild Cognitive Impairment

Claire André, Marc-André Bédard, Véronique Daneault, Rebekah Wickens, Jean-Paul Soucy, Dominique Lorrain, Célyne Bastien, Carol Hudon, Nicola Andrea Marchi, Jacques Montplaisir, Nadia Gosselin, Julie Carrier.

Supplementary Table 1. Clusters showing significantly higher cholinergic uptake in women compared to men.

Supplementary Figure 1. Associations between REM sleep characteristics and demographic variables.

Supplementary Table 2. Clusters significantly associated with REM sleep EEG slowing ratios.

Supplementary Figure 2. Fully adjusted associations between REM sleep EEG slowing ratio and cholinergic uptake.

Supplementary Table 3. Associations between the frontal resting-state wakefulness EEG slowing ratio and FEOBV-SUVr maps.

Supplementary Table 4. Clusters significantly associated with REM sleep EEG slowing ratios in sex-stratified analyses.

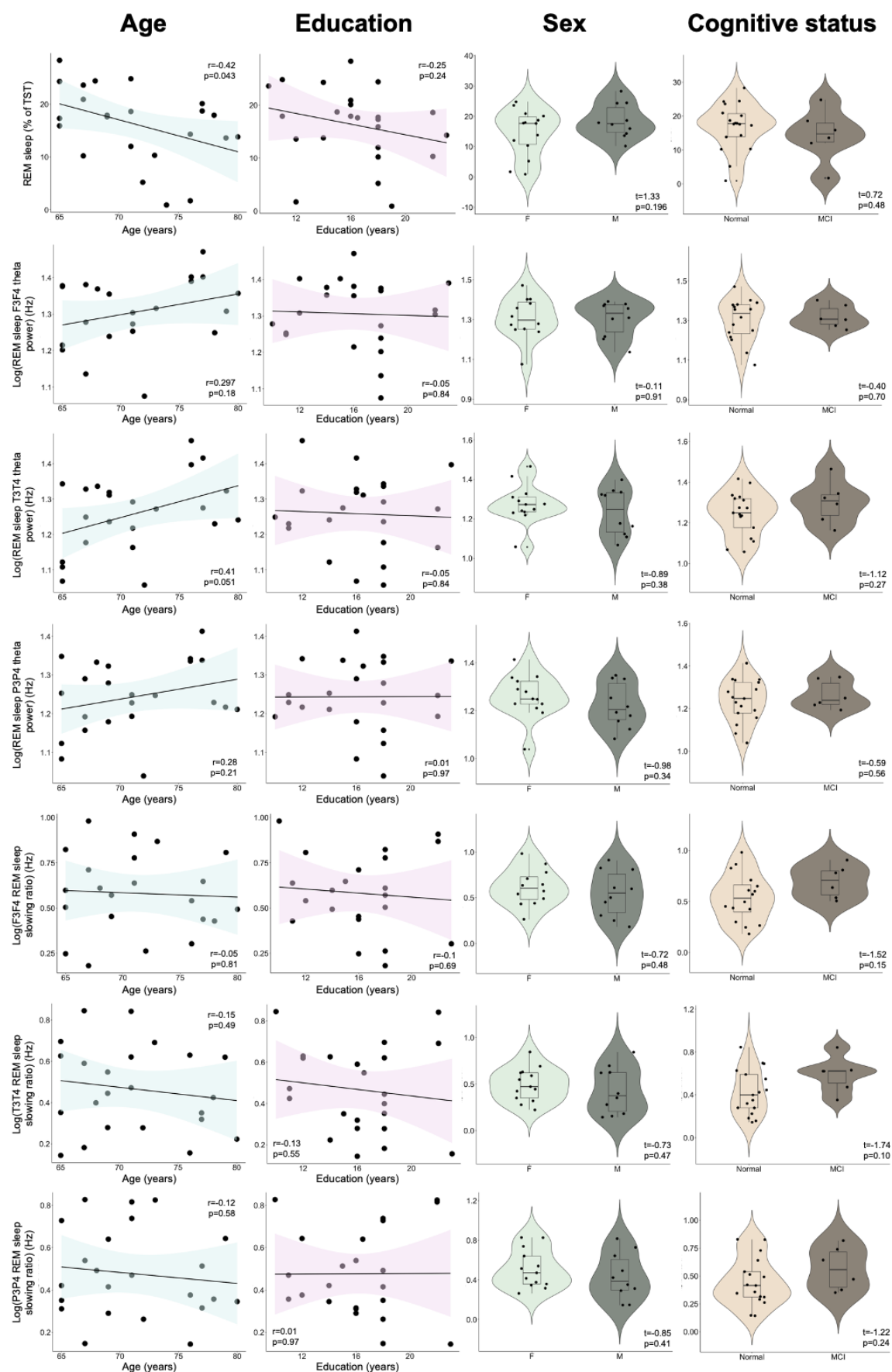
Supplementary Table 1. Clusters showing significantly higher cholinergic uptake in women compared to men.

Cluster label	k	T	Cluster-level	MNI coordinates		
			pFWE-corr	x	y	z
Unadjusted Two-sample analysis						
L hippocampus, middle temporal gyrus, superior temporal gyrus, supramarginal gyrus, Heschl gyrus, rolandic operculum, insula, superior temporal pole, putamen, basal forebrain, fusiform gyrus, lingual gyrus, parahippocampal gyrus, amygdala, thalamus, calcarine cortex, posterior cingulate cortex, precuneus	7603	5.31	<0.001	-32	-18	-12
R hippocampus, parahippocampal gyrus, amygdala	1691	4.99	0.18	28	-33	-3
R lingual gyrus, calcarine cortex, superior occipital gyrus	2577	4.86	0.048	14	-92	-8
L middle temporal gyrus, inferior temporal gyrus, angular gyrus	2768	4.28	0.036	-45	-74	20
R supramarginal gyrus, inferior parietal gyrus, angular gyrus	720	4.27	0.74	45	-38	39
R middle cingulate gyrus	807	4.25	0.67	18	-30	40
L middle occipital gyrus	477	4.25	0.91	-27	-82	8
R insula, putamen, superior temporal gyrus, middle temporal gyrus, inferior temporal gyrus, supramarginal gyrus, Rolandic operculum, postcentral gyrus, precentral gyrus	6777	4.24	<0.001	38	-16	0
L middle occipital gyrus, lingual gyrus, calcarine gyrus	1316	4.14	0.33	-10	-94	0
L middle temporal gyrus, superior temporal gyrus, middle temporal gyrus, temporal pole	1100	4.09	0.45	-48	-14	-16
L inferior frontal gyrus, middle frontal gyrus	1376	4.04	0.30	-52	24	8
R middle temporal gyrus	463	4.03	0.92	52	-66	-3
R lingual gyrus, precuneus, calcarine cortex, cuneus.	656	3.95	0.79	20	-54	0
B basal forebrain, olfactory cortex, gyrus rectus, superior frontal gyrus	1124	3.86	0.44	2	9	-10
L inferior parietal gyrus	633	3.82	0.81	-33	-50	44
L precentral gyrus	181	3.66	1.00	-21	-26	54
L postcentral gyrus	422	3.64	0.94	-44	-16	33
R inferior frontal gyrus, middle frontal gyrus	405	3.62	0.95	56	24	15
L inferior frontal gyrus	192	3.60	1.00	-40	9	26
B superior frontal gyrus, L anterior cingulate cortex	725	3.53	0.74	-14	62	-9
R lingual gyrus	137	3.46	1.00	21	-58	-4
R inferior frontal gyrus	107	3.42	1.00	44	12	27
R postcentral gyrus	116	3.35	1.00	32	-39	58
L middle frontal gyrus, L inferior frontal gyrus	209	3.27	1.00	-46	32	27
L inferior temporal gyrus	111	3.16	1.00	-57	-18	-28
Age-adjusted Two-sample analysis						
L hippocampus, parahippocampal gyrus, lingual gyrus, amygdala	1627	4.63	0.20	-32	-18	-14
L insula, Heschl gyrus, Rolandic operculum	766	4.52	0.70	-38	-15	15
R hippocampus, parahippocampal gyrus	856	4.43	0.63	32	-30	-10
R lingual gyrus, inferior occipital gyrus	368	4.12	0.96	14	-92	-8
B basal forebrain, olfactory gyrus	369	3.88	0.96	2	8	-9

R superior temporal gyrus, insula, Rolandic operculum.	910	3.73	0.59	57	-6	2
R supramarginal gyrus	149	3.66	1.00	58	-27	32
R superior temporal gyrus, middle temporal gyrus	309	3.58	0.98	57	-12	-4
L middle temporal gyrus	467	3.48	0.92	-56	-56	8
L superior temporal gyrus, Rolandic operculum, supramarginal gyrus	506	3.45	0.90	-57	-34	22
L superior temporal gyrus	241	3.42	0.99	-48	2	-12
L middle occipital gyrus, calcarine cortex	101	3.39	1.00	-10	-94	0
R superior temporal pole, amygdala, hippocampus	138	3.32	1.00	28	3	-22
L inferior frontal gyrus, middle frontal gyrus, insula	370	3.29	0.96	-44	42	-3

Results of two-sample voxel-wise comparisons showing higher cholinergic uptake in women ($n=14$) compared to men ($n=10$), at the $P < 0.005$ uncorrected level combined with a minimum cluster size of $k=100$. Clusters surviving a cluster-level FWE correction are highlighted in bold. For each cluster, upper coordinates correspond to the statistical peak of the cluster. Abbreviations: FWE, family wise error; L, left; MNI, Montreal Neurological Institute; Nb, number; R, right.

Supplementary Figure 1. Associations between REM sleep characteristics and demographic variables.



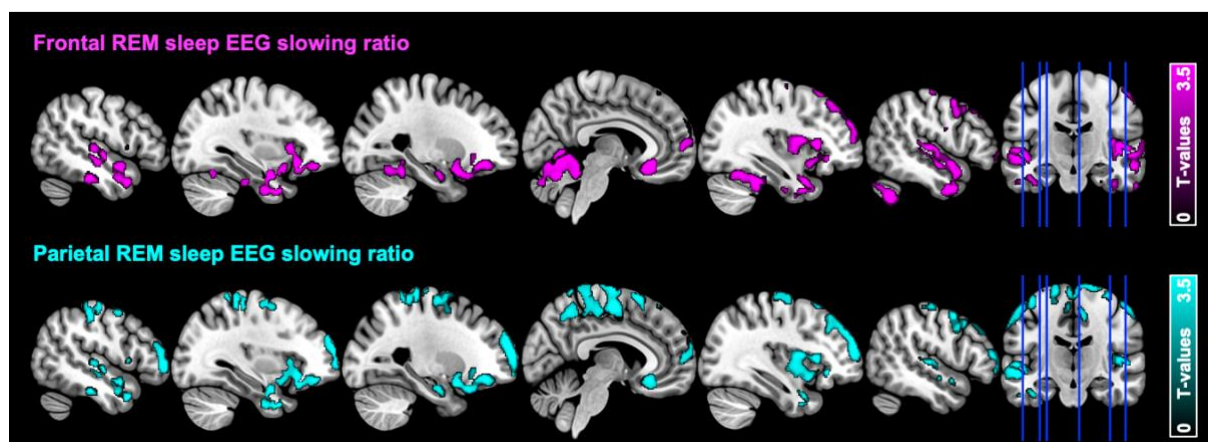
Associations between REM sleep percentage, theta power and EEG slowing ratios and demographic variables. Between-group differences in REM sleep outcomes according to sex or cognitive status were assessed using Student T-tests, and associations with age and education are assessed using Pearson's correlations. No association survived a false discovery rate correction for multiple comparisons. Abbreviations: Hz, Hertz; REM, rapid eye movement; TST, total sleep time.

Supplementary Table 2. Clusters significantly associated with REM sleep EEG slowing ratios.

Cluster label	k	T-value	p _{FWE-corrected}	MNI coordinates		
				x	y	z
Frontal REM sleep EEG slowing ratio (n=22)						
B vermis, R lingual gyrus, B fusiform gyrus, B middle temporal gyrus, B superior temporal gyrus, B inferior temporal gyrus, B temporal pole, R parahippocampal gyrus, B hippocampus, B amygdala, B insula, R postcentral gyrus, R precentral gyrus, R supplementary motor area, R superior frontal gyrus, B middle frontal gyrus, R inferior frontal gyrus pars orbitalis, B superior frontal gyrus-medial orbital, R inferior frontal gyrus-opercular part, R inferior frontal gyrus-triangular part, B anterior cingulate cortex, B rectus gyrus, B medial orbital gyrus, B posterior orbital gyrus, L anterior orbital gyrus.	38622	5.58	<0.001	6	-58	-6
B cuneus, L precuneus, L angular gyrus, L middle and superior occipital gyrus, L calcarine cortex.	2667	5.51	0.014	-6	-78	28
B precuneus, R superior parietal gyrus, L inferior parietal gyrus, B middle cingulate gyrus, B postcentral gyrus, L precentral gyrus, B paracentral lobule, B supplementary motor area, L superior frontal gyrus, L middle frontal gyrus.	8201	4.84	<0.001	-14	-46	72
L superior frontal gyrus, L middle frontal gyrus, L inferior frontal gyrus-triangular part.	5451	4.48	<0.001	-2	45	45
Parietal REM sleep EEG slowing ratio (n=23)						
B cuneus, B precuneus, B superior occipital gyrus, L middle occipital gyrus, B fusiform gyrus, B postcentral gyrus, B superior parietal gyrus, L inferior parietal gyrus, B precentral, B superior frontal gyrus, B middle frontal gyrus, B inferior frontal gyrus-opercular part, B inferior frontal gyrus-triangular part, B inferior frontal gyrus pars orbitalis, R Rolandic operculum, B supplementary motor area, B olfactory cortex, B superior frontal gyrus-medial and medial orbital parts, B rectus gyrus, B orbital gyrus (anterior, medial and posterior parts), R anterior cingulate cortex, B insula, B middle cingulate gyrus, L hippocampus, L amygdala, B paracentral lobule, L putamen, R Heschls gyrus, B superior temporal gyrus, B temporal pole, B middle temporal gyrus, B inferior temporal gyrus, B cerebellar hemispheres.	51051	4.25	<0.001	-6	-78	28
B vermis, R fusiform gyrus, R lingual gyrus, R parahippocampal gyrus.	2170	3.72	0.039	4	-60	-6

Whole-brain voxel-wise negative multiple regression analyses were performed between REM sleep EEG slowing ratios on frontal (averaged F3-F4) and parietal (averaged P3-P4) derivations, and FEOBV SUVR maps. Results are presented at the $P < 0.005$ (uncorrected) level combined with a FWE cluster-level correction, controlling for sex. For each cluster, upper coordinates correspond to the statistical peak of the cluster. Abbreviations: B, bilateral; FWE, family wise error; L, left; MNI, Montreal Neurological Institute; R, right; REM, rapid eye movement.

Supplementary Figure 2. Fully adjusted associations between REM sleep EEG slowing ratio and cholinergic uptake.



Whole-brain negative voxel-wise regressions between REM sleep EEG slowing ratio on F3-F4 (top, $n=22$) and P3-P4 (bottom, $n=23$) derivations and FEOBV-SUVR maps corrected for partial volume effects, controlling for age, sex and cognitive status. Results are presented at the $P < 0.005$ uncorrected level, combined with a cluster-level FWE correction. Abbreviations: FEOBV, [^{18}F]-Fluoroethoxybenzovesamicol; FWE, family wise error; SUVR, standard uptake value ratio, REM, rapid eye movement.

Supplementary Table 3. Association between the frontal resting-state wakefulness EEG slowing ratio and FEOBV-SUVR maps.

Cluster label	k	T-value	p _{FWE-corrected}	MNI coordinates		
				x	y	z
Right superior temporal pole	107	4.38	1.000	36	9	-27
Left superior frontal gyrus	193	3.98	0.997	-26	15	51
Left superior frontal gyrus	137	3.79	0.999	-26	39	44

Whole-brain voxel-wise negative multiple regression between the resting-state wakefulness EEG slowing ratio computed on frontal derivations (averaged F3-F4, $n=18$) and FEOBV-SUVR maps, controlling for sex. Results are presented at the $P < 0.005$ (uncorrected) level, combined with a minimum cluster size of $k=100$. Abbreviations: FEOBV, [^{18}F]-Fluoroethoxybenzovesamicol; FWE, family wise error; SUV, standard uptake value ratio; REM, rapid eye movement.

Supplementary Table 4. Clusters significantly associated with REM sleep EEG slowing ratios in sex-stratified analyses.

Group	Cluster label	k	T-value	p _{FWE-corrected}	MNI coordinates		
					x	y	z
Frontal REM sleep EEG slowing ratio (n=22)							
Women	Cerebellar vermis	220	7.80	0.995	6	-40	-21
	R hippocampus, parahippocampal gyrus, putamen, amygdala, basal forebrain, nucleus accumbens, gyrus rectus, medial orbital gyrus	3651	1.36	0.003	27	-20	-15
	L insula, superior temporal gyrus, inferior temporal gyrus, middle temporal gyrus, amygdala, hippocampus, putamen, fusiform gyrus, basal forebrain, gyrus rectus, medial orbital gyrus	5307	6.88	<0.001	-39	-9	-15
	L anterior cingulate cortex	142	4.55	0.999	-2	38	6
	L superior frontal gyrus	133	4.41	1.000	-24	58	4
	R superior frontal gyrus	101	4.16	1.000	22	44	38
Men	B lingual gyrus	582	8.98	0.296	6	-62	-2
	L precentral gyrus, supplementary motor area, precuneus, paracentral lobule, superior parietal gyrus, postcentral gyrus	458	8.80	0.511	-22	-22	72
	L postcentral gyrus, precentral gyrus, paracentral lobule	624	8.27	0.243	-33	-30	58
	R superior frontal gyrus, precentral gyrus	2140	7.84	<0.001	27	27	56
	R middle frontal gyrus	322	7.12	0.816	42	8	56
	L superior frontal gyrus	293	7.06	0.873	-14	38	54
	R precentral gyrus, postcentral gyrus	180	6.83	0.992	63	-2	27
	R superior parietal gyrus, precuneus, paracentral lobule	818	6.74	0.097	15	-63	68
	L postcentral gyrus, supramarginal gyrus	294	6.63	0.871	-57	-14	46
	L supplementary motor area, paracentral lobule, middle cingulate cortex	182	6.53	0.991	-10	-22	50
	R superior frontal gyrus	169	6.23	0.995	27	62	15
	L orbitofrontal cortex	221	6.02	0.969	-27	39	-14
	L middle frontal gyrus	242	5.81	0.949	-50	21	38
	L middle temporal gyrus, angular gyrus, superior temporal gyrus	754	5.71	0.132	-64	-32	3
	R superior frontal gyrus	233	5.68	0.958	20	66	-8
	R superior frontal gyrus	191	5.62	0.988	8	56	-2
	L precuneus, superior parietal gyrus	243	5.59	0.948	-33	24	51
	L precuneus, superior parietal gyrus, paracentral lobule	447	5.59	0.534	-4	-42	74
	L supplementary motor area, L superior frontal gyrus	113	5.56	1.000	-12	22	62
	R supplementary motor area	102	5.45	1.000	6	18	66
	R fusiform gyrus	237	5.43	0.954	30	-33	-26
	L middle occipital gyrus, inferior occipital gyrus	152	5.42	0.998	-28	-94	-2
	L precentral gyrus, middle frontal gyrus, postcentral gyrus	359	5.30	0.734	-42	-6	62
	R supplementary motor area	123	5.26	1.000	8	-6	57
	L orbitofrontal cortex	110	5.20	1.000	-22	15	-22
	L superior frontal gyrus, middle frontal gyrus	586	5.17	0.290	-28	56	18
	R fusiform gyrus	116	5.10	1.000	39	-57	-20
	R cuneus	159	5.07	0.997	14	-102	9
	R orbitofrontal cortex	104	4.91	1.000	26	20	-18

	R middle temporal gyrus, inferior temporal gyrus, superior temporal gyrus	715	4.87	0.158	66	-20	-21
	L rectus gyrus	117	4.77	1.000	-6	26	-18
	R superior temporal gyrus	104	4.72	1.000	64	-33	4
	L superior occipital gyrus	211	4.66	0.997	-21	-86	36
	L inferior temporal gyrus	100	4.64	1.000	-50	-60	-16
	R superior occipital gyrus, cuneus	159	4.49	0.997	22	-88	28
	B cuneus	102	4.39	1.000	4	-84	26
	L inferior temporal gyrus	134	4.29	0.999	-45	-42	-27
	R lingual gyrus, fusiform gyrus	103	4.17	1.000	20	-50	-9
Parietal REM sleep EEG slowing ratio (n=23)							
Women	L inferior temporal gyrus, insula, hippocampus, amygdala, parahippocampal gyrus, fusiform, inferior temporal gyrus, temporal pole	5426	6.42	<0.001	-39	-20	-22
	Cerebellar vermis	135	5.17	0.999	3	-40	-20
	R insula, putamen, hippocampus, amygdala, parahippocampal gyrus	1719	4.90	0.097	36	-3	-10
	R thalamus, nucleus accumbens, basal forebrain, caudate nucleus, gyrus rectus, medial orbital gyrus	703	4.23	0.664	12	-4	12
	L superior frontal gyrus	319	4.21	0.972	-26	58	4
	L middle frontal gyrus	103	3.87	1.000	-26	24	45
Men	L precentral gyrus, postcentral gyrus	850	10.45	0.136	-34	-30	60
	R superior frontal gyrus, precentral gyrus, supplementary motor area	816	8.90	0.156	20	-15	75
	L postcentral gyrus, supramarginal gyrus	276	3.83	0.936	-58	-15	45
	L precentral gyrus, paracentral lobule, supplementary motor area	407	6.63	0.718	-21	-22	74
	R superior frontal gyrus, middle frontal gyrus	395	6.37	0.742	26	63	15
	L superior frontal gyrus	187	3.55	0.993	-8	38	57
	L superior frontal gyrus	108	3.54	1.000	27	28	56
	R supplementary motor area	181	3.53	0.995	9	-6	58
	L precuneus, paracentral lobule, superior parietal gyrus	417	6.18	0.698	-4	-40	72
	L supplementary motor area, paracentral lobule, middle cingulate gyrus	148	6.11	0.999	-10	-22	50
	L middle frontal gyrus, inferior frontal gyrus	398	6.09	0.736	-34	26	51
	L precentral gyrus, middle frontal gyrus	296	5.98	0.912	-50	-2	52
	L superior frontal	130	5.92	1.000	-21	2	70
	R middle frontal gyrus, inferior frontal gyrus	470	5.75	0.592	44	28	38
	R superior parietal gyrus, precuneus	120	5.65	1.000	15	-63	68
	R middle frontal gyrus	183	5.59	0.994	38	6	60
	L superior frontal gyrus	322	5.32	0.874	-22	63	9
	B lingual gyrus, cerebellar vermis	333	5.22	0.856	6	-60	-2
	L middle temporal gyrus, superior temporal gyrus	234	5.18	0.973	-66	-30	2
	L superior frontal gyrus	100	5.15	1.000	-15	63	20
	R superior frontal gyrus	136	5.03	0.999	9	54	-2
	L superior occipital gyrus, middle occipital gyrus	317	4.54	0.882	-21	-87	36
	L middle frontal gyrus, orbitofrontal cortex	114	4.50	1.000	-27	38	-12
Temporal REM sleep EEG slowing ratio (n=23)							
Women	Cerebellar vermis	287	7.26	0.985	8	-44	-26
	L anterior cingulate cortex, nucleus accumbens, basal forebrain, putamen, caudate nucleus	387	4.41	0.953	-6	22	-6
	R caudate nucleus	173	4.37	0.998	14	21	-8

	R superior temporal pole	105	4.06	1.000	50	4	-12
	R putamen	282	3.70	0.986	30	18	3
Men	Cerebellar vermis, B lingual gyrus, R calcarine cortex	595	7.82	0.387	4	-63	-3
	L postcentral gyrus, supramarginal gyrus, precentral gyrus	303	7.49	0.906	-57	-12	45
	L postcentral gyrus, precentral gyrus	451	6.83	0.639	-28	-34	68
	L middle frontal gyrus, inferior frontal gyrus	468	6.70	0.606	-51	20	36
	R middle frontal gyrus, superior frontal gyrus, inferior frontal gyrus	650	6.22	0.313	42	28	33
	R postcentral gyrus	106	6.18	1.000	63	-2	28
	R superior frontal gyrus, middle frontal gyrus	559	6.17	0.442	24	4	62
	R precentral gyrus	117	6.13	1.000	42	-22	64
	R superior frontal gyrus	136	5.90	0.999	16	36	54
	R superior frontal gyrus, middle frontal gyrus	104	5.77	1.000	26	50	38
	L precentral gyrus, paracentral lobule	153	5.61	0.999	-26	-24	68
	R inferior parietal gyrus	142	5.55	0.999	48	-45	56
	L superior frontal gyrus	137	5.48	0.999	-27	58	12
	R precuneus	122	5.45	1.000	8	-50	72
	R precentral gyrus, superior frontal gyrus	314	5.02	0.890	26	-20	72
	R inferior frontal gyrus, insula	362	4.39	0.811	44	26	3

Whole-brain voxel-wise negative multiple regression analyses were performed between REM sleep EEG slowing ratios on frontal (averaged F3-F4), temporal (averaged T3-T4) and parietal (averaged P3-P4) derivations and FEOBV-SUVr maps in groups stratified by sex. Results are presented at the $P < 0.005$ (uncorrected) level combined with a minimum cluster size of $k=100$, controlling for age. Clusters surviving a cluster-level FWE correction are highlighted in bold. For each cluster, coordinates correspond to the statistical peak of the cluster. Abbreviations: B, bilateral; FWE, family wise error; L, left; MNI, Montreal Neurological Institute; R, right; REM, rapid eye movement.